

Run Wang

runw@clemson.edu | +1 (518) 961-8733 | Clemson, SC | runwang123.github.io

EDUCATION

Clemson University

Ph.D. Student in Computer Science

– Concentration: 3D Computer Vision, AI Security and Mobile Computing

Clemson, SC

Aug. 2025 – Present

Carnegie Mellon University

M.S. in Electrical and Computer Engineering (Applied Advanced Program)

Pittsburgh, PA

Sep. 2021 – May 2023

Rensselaer Polytechnic Institute

B.S. in Computer & Systems Engineering

Troy, NY

Sep. 2017 – May 2021

SKILLS

Programming: C, C++, C#, Python (PyTorch), Java

Tools & Frameworks: Docker, Kubernetes, Terraform, Hugging Face

Research Areas: Deep Learning, Autonomous Driving, Mobile Computing

PUBLICATIONS AND ACKNOWLEDGEMENTS

R. Wang et al., “FlexMap: Generalized HD Map Construction from Flexible Camera Configurations.”
Preprint.

Z. Wang, M. Kim, **R. Wang et al.**, “Design Diagnosis Using LLMs: A Case Study of Recreational Fishing Choice Experiments”
Under Review.

C. Zhou, **R. Wang et al.**, “FF3R: Feedforward Feature 3D Reconstruction from Unconstrained Views.”
CVPR Findings, 2026.

R. Wang et al., “Lightweight Detection of Abnormal Battery Drain Induced by Network Operations of Mobile Apps.”
IEEE INFOCOM, 2026. [Acceptance Rate: 329/1740 \(18.91%\)](#)

R. Wang et al., “MAPP: Predictive UI View Pre-caching for Improving the Responsiveness of Mobile Apps.”
IEEE/ACM IWQoS, 2025. [Acceptance Rate: 81/326 \(24.8%\)](#)

W. Zhang et al., “Component Segmentation of Engineering Drawings Using Graph Convolutional Networks.”
Computers in Industry, vol. 147, Art. no. 103885, 2023. (Acknowledged Contributor)

RESEARCH EXPERIENCE

Graduate Research Assistant

Clemson University

- Supervisors: Dr. Siyu Huang and Dr. Mert D. Pesé
- Conducting research on 3D computer vision and trustworthy AI for autonomous driving systems.

Aug. 2025 – Present

Clemson, SC

Graduate Research Associate

The Ohio State University

- Supervisor: Prof. Xiaorui Wang
- Designed machine learning models and time-series algorithms for mobile and edge performance optimization.
- Two papers are accepted by IEEE IWQoS 2025 and IEEE INFOCOM 2026.

May 2024 – Aug. 2025

Columbus, OH

Research Assistant

CERLAB, Carnegie Mellon University

- Supervisors: Prof. Kenji Shimada and Dr. Wentai Zhang
- Developed deep learning methods for component extraction from engineering drawings.
- Adapted PSPNet, DeepLabV3+, and SketchGNN, achieving 84% overall accuracy.

May 2022 – May 2023

Pittsburgh, PA

Research Assistant

CyLab, Carnegie Mellon University

- Supervisor: Dr. Cai Yang

Jan. 2022 – May 2022

Pittsburgh, PA

- Collected and analyzed 80+ Linux-based malware samples via dynamic system-level tracing.
- Constructed an in-house behavioral dataset with 14 discriminative features.

Research Assistant

Jan. 2020 – Dec. 2020

RPI-IBM Cognitive and Immersive Systems Lab

Troy, NY

- Supervisors: Dr. Jeff Kephart and Dr. Xiangyang Mou
- Implemented a real-time gesture recognition system using C# and Python with Intel RealSense cameras.
- Collected 600+ gesture samples and extracted skeletal motion features.

INDUSTRY EXPERIENCE

Application Developer – Full Stack

May 2023 – Apr. 2024

Computer Packages Inc.

Rockville, MD

- Supported migration from legacy systems to modern architectures while ensuring data integrity.
- Optimized SQL Server queries using stored procedures and execution plan tuning.

PROJECT EXPERIENCE

Cosmospapers Research Assistant Platform

Jan. 2026 – Present

Demo - [Github](#) - [Github Star: 23](#)

- Developed RAG-based paper search engine.
- Collected, vectorized 9000+ papers from 10+ top AI and Security conferences.
- Designed in-house algorithms to cluster and extract keywords of research trends.

User Recommendation Microservice

Aug. 2022 – Dec. 2022

GCP, Kubernetes, Spark, FastAPI, Vert.x

- Developed Spark-based ETL pipelines to preprocess ~1TB Twitter dataset on GCP Dataproc.
- Orchestrated AWS infrastructure and achieved 10,000 RPS within \$0.7/hr budget.